

[HCI/2024/Q4]

4 Name your Jupyter Notebook as

TASK4_<your name>_<centre number>_<index number>.ipynb

An airline company currently keeps paper records about their customers, the daily flights and the tickets sold. The manager wants to create a suitable database to store the data.

The database TRIP will have the following tables:

Flight:

- FlightNo – unique code for the flight, for example TR425
- DepartCity – the departure city of the flight
- ArrivalCity – the arrival city of the flight
- DepartTime – the departure time of the flight, for example 2220
- ArrivalTime – the arrival time of the flight, for example 2335

Customer:

- CustomerNo – unique integer number per customer, for example 12
- Name – the name of the customer
- Gender – the gender of the customer
- DOB – the date of birth of the customer, for example 21041997 means 21 Apr 1997

Ticket:

- TicketNo – unique integer number per ticket, for example 1010
- CustomerNo – the customer's unique number
- FlightNo – the unique flight code of the ticket
- Date – the date of flight on the ticket
- Seat – the seat category of the ticket, for example Economy

For each of the sub-tasks, 4.1 to 4.3, add a comment statement at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

In [1]:

```
#Task 4.1  
Program Code
```

Output:

Task 4.1

Write a Python program that uses SQL code to create the database TRIP with the three tables given. Define the primary and foreign keys for each table.

[4]

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Task 4.2

The text files `FLIGHT.txt`, `CUSTOMER.txt` and `TICKET.txt` store the comma-separated values for each of the tables in the database. Write a Python program to read in the data from each file and store the data in the correct place in the database. [3]

Task 4.3

The company has rescheduled flight TR808's departure time to 18:15 and arrival time to 00:15. Write a Python program to update the database with this information, then display all the flights departing from Singapore in ascending order of departure time.

A customer named Shao Xingjuan wants to cancel her tickets numbered 1023 and 1024. Write a Python program to update the database with this cancellation. [7]

Save your Jupyter Notebook for Task 4.

Task 4.4

The company manager wants to filter the ticket data by `Date` and display the results in a web browser. Write a Python program and the necessary files to create a web application that:

- Receives a `Date` string from a HTML form,
- Returns an HTML document that enables the web browser to display a table tabulating the customer's name, departure city, arrival city and the seat category for tickets sold on that date.

Save your Python program as:

`TASK4_4_<your name>_<centre number>_<index number>.py`

with any additional files/subfolders in a folder named:

`TASK4_4_<your name>_<centre number>_<index number>`

Run the web application with the date entered as '08082024'.

Save the web page output as:

`TASK4_4_<your name>_<centre number>_<index number>.html` [13]